Breadfast

Task 1 Report

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Introduction

To reduce waste of fresh and baked items, Breadfast has introduced a new type of discount called "Automated Discounts". This type of discount mainly targets products or categories approaching OOV (out of validity) dates.

Breadfast backend engine is built based on configured rules, where the closer we approach the OOV date, the more aggressive/higher the automated discount will become.

Once configurations are set, Breadfast run the discounts, to reflect in products on the shop catalog.

You can explore the dashboard for more insights here.

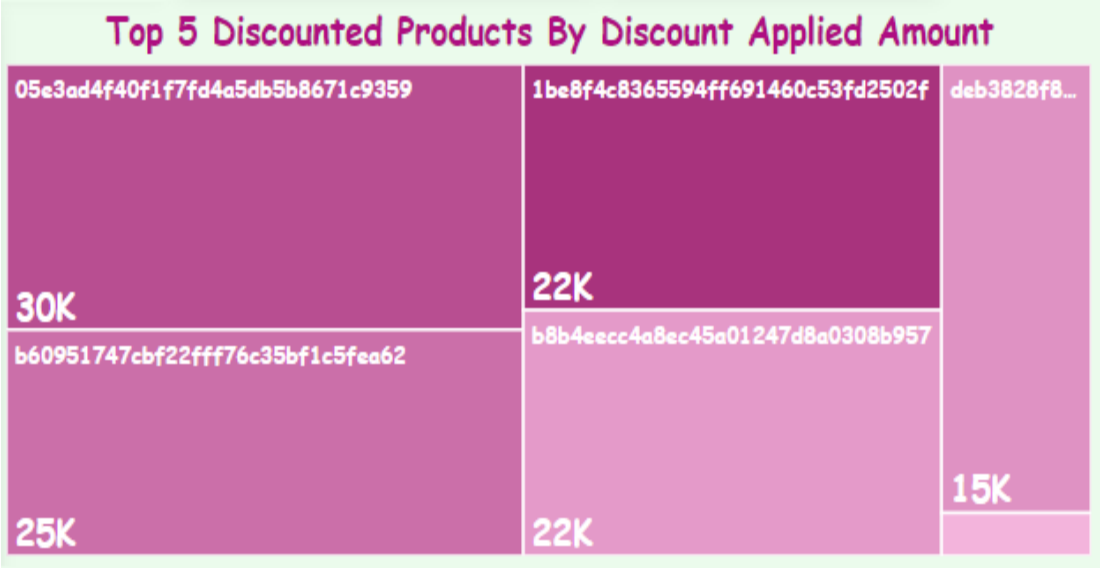
This report aims to analyze the results based on the data provided from the dashboard.

Key Insights

1. **Discount Monitoring Insights**
   - **The total value of the discounts applied was 961.55K.**
   - **Sum of Products' Original Price with a Discount are 4.08M.**
   - **Net Percentage After Discount is 76.46%.**
2. **Volume of Products Sold and OOV Insights**
   - **A total of 104.39K units were sold with automated discounts.**
   - **Total OOV (Out of Validity) products were 251.55K units.**
   - **Total waste after discounts amounted to 147.16K units**
   - **Percentage of OOV products remaining after discounts: 59%.**
3. **Top Discounted Products**

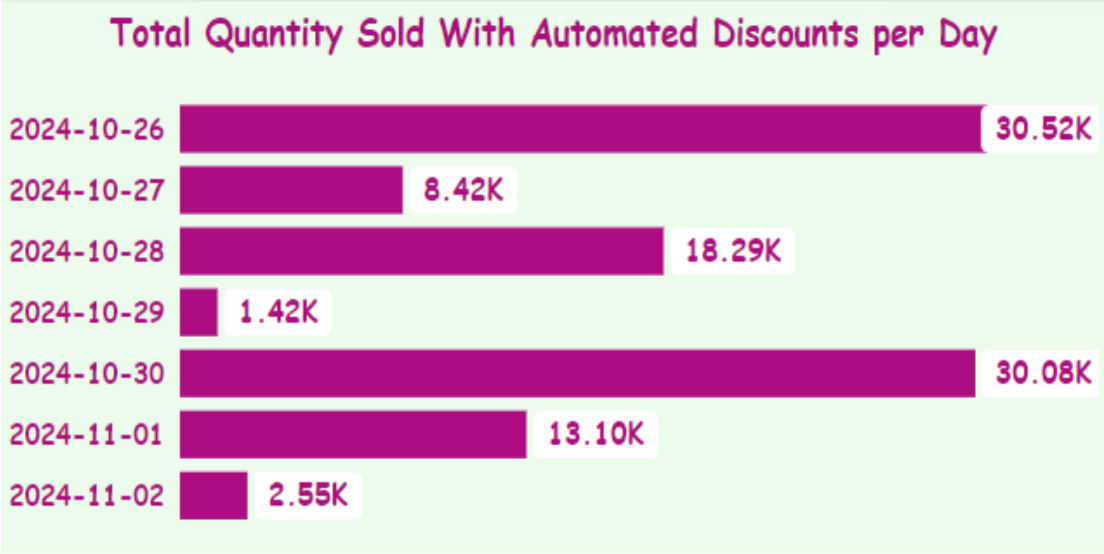
   The top five discounted products by discount amount were:

   - **Product 1: 30K.**
   - **Product 2: 25K.**
   - **Product 3: 22K**
   - **Product 4: 22K**
   - **Product 5: 15K**

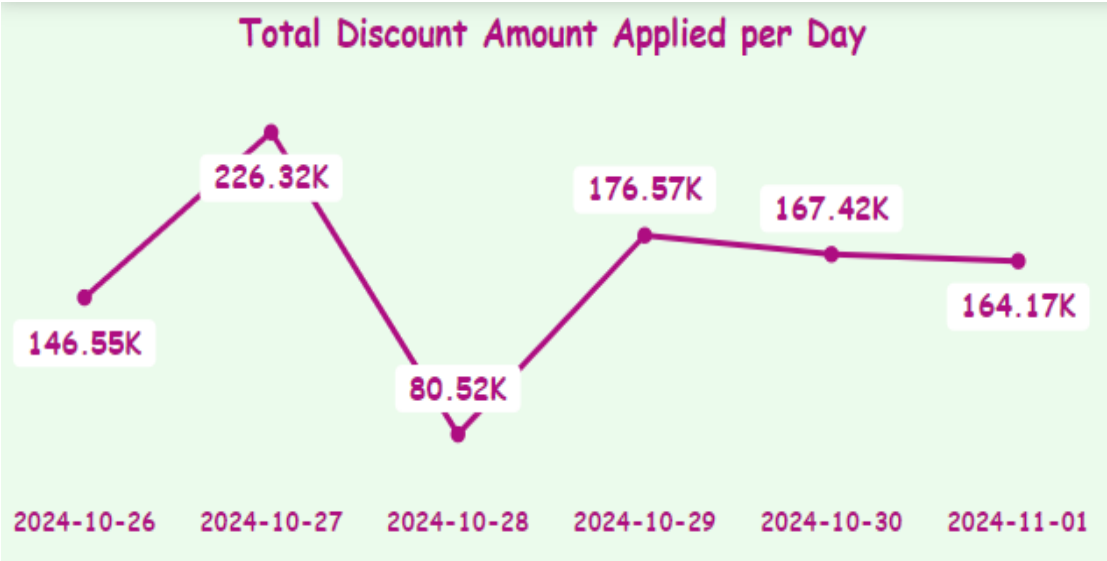


4. **Daily Insights:**

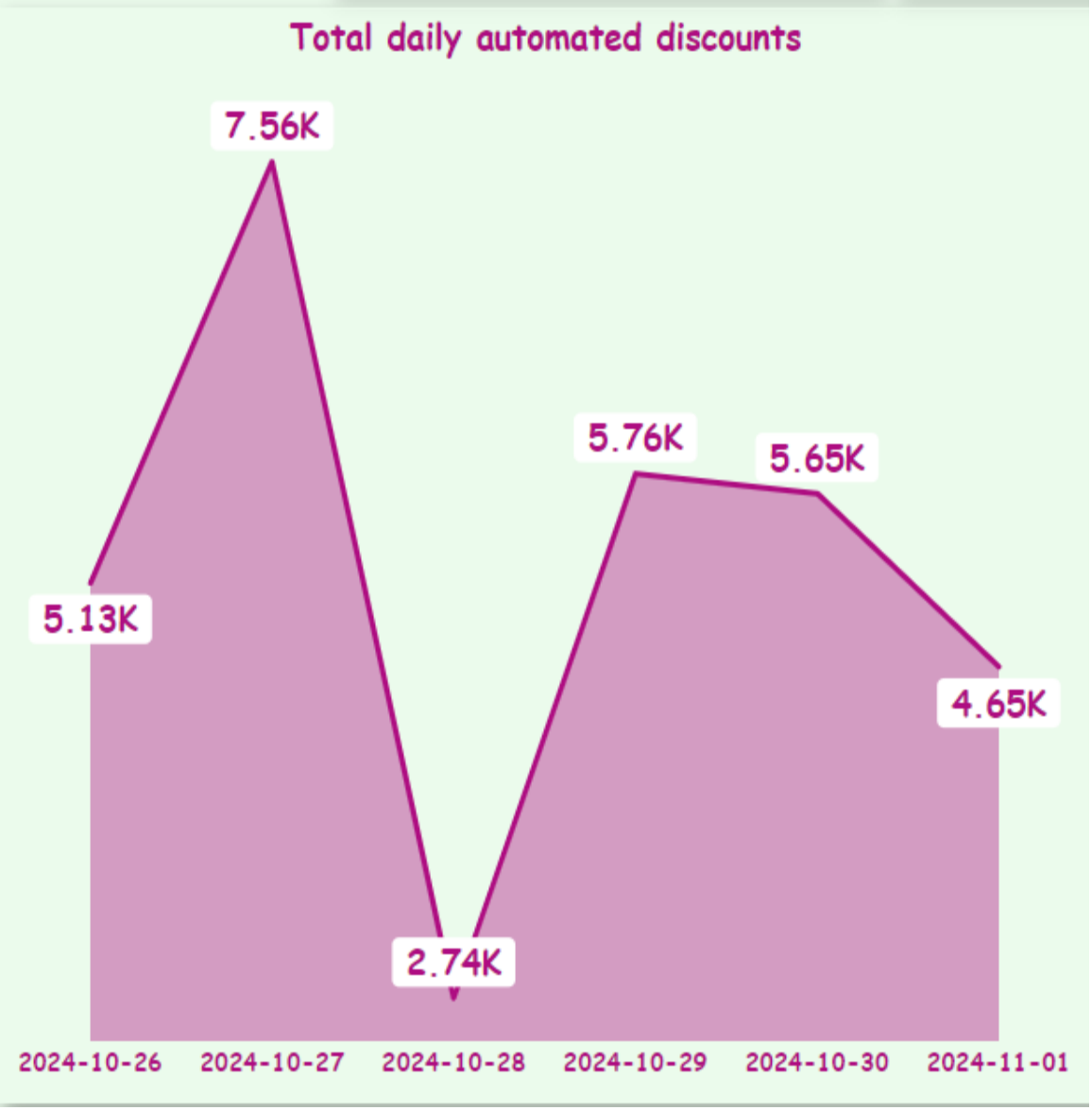
- **Total Quantity Sold with Automated Discounts per Day: This chart illustrates the daily sales volume of products sold with automated discounts. The data highlights fluctuations in sales, with peak days reaching up to 30.52K units. This trend indicates the impact of discounts on customer purchasing behavior.**



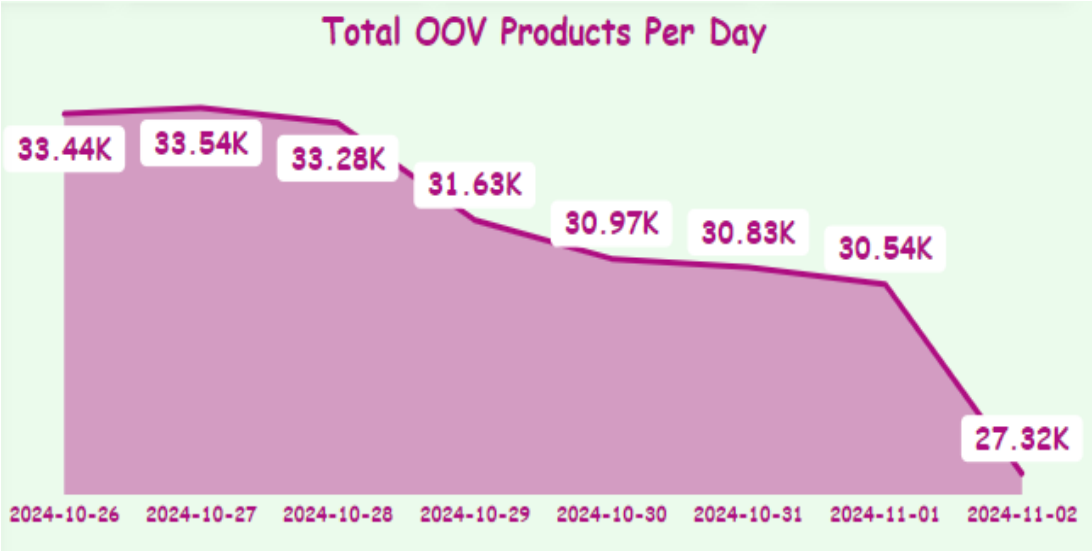
- **Total Discount Amount Applied per Day: This chart shows the total value of discounts applied daily. The range of discounts (80.52K to 226.32K) demonstrates the variability in discount application and the effectiveness of targeting specific products or categories.**



- **Total Daily Automated Discounts: This visualization tracks the number of automated discounts applied each day, ranging from 2.74K to 7.56K.**



- **Total OOV Products Per Day: This chart represents the daily count of products nearing expiration. The consistent range (27.32K to 33.54K) indicates stable inventory management practices but also highlights the need for improved discount strategies to reduce OOV counts.**



5. **Discount Termination:**
   - **Manually canceled: 30.27K units.**
   - **Exhausted stock: 1.20K units.**

Analysis Result

1 .Discount Spending Through Automated Discounts

- **The total discount spending during the period was 961.55K, reflecting the company's investment in reducing waste and encouraging purchases of products nearing expiration. The daily discount application varied significantly, ranging from 80.52K to 226.32K, highlighting fluctuations in the intensity of discount campaigns. These variations likely correspond to changes in product categories, customer demand, or inventory nearing its out-of-validity (OOV) dates .**

2 .Inventory Sold at a Discounted Price in Terms of Volume and Value

- **Breadfast successfully sold a total of 104.39K units under automated discounts, with the original price value of these products summing up to 4.08M. After applying discounts, the net price percentage dropped to 76.46%, indicating significant savings passed on to customers. These figures emphasize the program's success in driving sales for discounted items, suggesting that customers are responding positively to price reductions. However, careful analysis is needed to ensure these discounts are generating sufficient profit margins to justify their use.**

3 .Percentage of Out-of-Validity Products Remaining After Discounts

- **Despite the automated discounts, 59% of the products nearing expiration remained unsold, equating to 147.16K units out of a total of 251.55K OOV products. This indicates that while the discounting strategy is effective to some extent, there is a significant portion of inventory still wasted. The high percentage of leftover OOV products suggests the need for more aggressive or targeted discounting strategies, potentially coupled with better forecasting and demand planning to minimize inventory waste.**

4 .Top Discounted Products

- **The top five discounted products accounted for substantial savings, with discounts ranging from 30K for the highest product to 15K for the fifth. These products likely represent either fast-moving items or those with the highest waste risk. Understanding which products are repeatedly discounted can help refine the inventory management process and improve stock planning to reduce reliance on discounts. This data also provides a starting point for analyzing the profitability and success of discount campaigns for specific items.**

5 .Impact on Sales and Waste

- **The discounting strategy had a noticeable impact on sales, with daily quantities sold peaking at 30.52K units. However, sales volumes fluctuated significantly across the observed period, suggesting that customer response may vary based on the product mix or discount depth. On the waste side, despite these efforts, 59% of OOV products remained unsold, underscoring the challenge of minimizing waste entirely. This highlights the need to balance discount aggressiveness with profitability and explore additional measures, such as dynamic pricing or bundling strategies, to maximize sales and reduce waste.**

6 .Discount Termination

- **The reasons for discount termination provide further context for inventory outcomes. A significant portion of discounts (30.27K units) was manually canceled, possibly due to operational decisions or stock adjustments. Additionally, 1.20K units were sold out, showing the effectiveness of some discount campaigns in completely depleting inventory. The number of daily discounts applied ranged from 2.74K to 7.56K, reflecting variability in inventory turnover and discount strategy execution. These metrics can guide future optimization of automated discounts and operational efficiencies.**

Conclusion

The automated discounting strategy implemented by Breadfast has demonstrated its potential in reducing waste and driving sales, as evidenced by the 104.39K units sold and a total discount spending of 961.55K. Despite this, significant challenges remain, with 59% of out-of-validity (OOV) products still unsold, resulting in 147.16K units wasted. The top discounted products and daily sales data indicate that customers are responsive to price reductions, but there is room for improvement in targeting and execution. To enhance the effectiveness of the discount program, Breadfast could consider adopting a more dynamic and data-driven approach. For example, refining discount rules based on product-specific demand patterns, introducing tiered discounts to further incentivize purchases of slower-moving inventory, and employing better forecasting methods to minimize OOV stock. Additionally, increasing customer awareness of discounted products through more prominent catalog placement or targeted marketing campaigns could improve sales further. By addressing these areas, Breadfast can maximize the efficiency of its discount program, reduce waste, and improve overall profitability.